

ASSIGNMENT - V

V. Multiplication by 's', Division by 's', Second Shifting Property :

1. Find $L^{-1} \left[\frac{s}{(s^2+a^2)^2} \right]$

Ans : $f(t) = \frac{t \sin at}{2a}$

2. Find $L^{-1} \left[\frac{s^2}{(s^2+a^2)^2} \right]$

Ans : $f(t) = \frac{\sin at + at \cos at}{2a}$

3. Find $L^{-1} \left[\frac{1}{s^2(s^2+a^2)} \right]$

Ans : $f(t) = \frac{1}{a^2} \left(t - \frac{\sin at}{a} \right)$

4. Find $L^{-1} \left[\frac{1}{s^3(s^2+1)} \right]$

Ans : $f(t) = \frac{t^2}{2} + \cos t - 1$

5. Find $L^{-1} \left[\frac{5-3e^{-3s}-2e^{-7s}}{s} \right]$

Ans : $f(t) = \begin{cases} 5, & 0 < t < 3 \\ 2, & 3 < t < 7 \\ 0, & t > 7 \end{cases}$

6. Find $L^{-1} \left[\frac{(s+1)e^{-\pi s}}{s^2+s+1} \right]$

Ans : $e^{-\frac{1}{2}(t-\pi)} \left\{ \cos \frac{\sqrt{3}}{2} (t-\pi) u(t-\pi) + \frac{1}{\sqrt{3}} \sin \frac{\sqrt{3}}{2} (t-\pi) u(t-\pi) \right\}$